Summarizing the predominant cover type of a watershed using a supervised image classification in Tenmile Creek, Oregon – A learning experience and review for future practices and applications

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Contents

Abstract 3
Introduction 4
Methods 5
Results 5
Discussion 9
Conclusion 10
References 10

Abstract

Supervised image classifications can produce an abundance of spatial and temporal information that can be used in ecological studies and analyses for a multitude of purposes. When conducting a classification, pixels within an image are into assigned classes based on their spectral signatures. To improve my spatial analysis skills and to gain information on the characteristics of the landscape within Tenmile Creek, I chose to conduct a supervised image classification within the watershed boundary of Tenmile Creek, Oregon. Several attempts at generating a supervised image classification were made, with many changes and adaptations added to each classification, resulting in a map of Tenmile Creek that distinguishes conifers of various successional periods, deciduous, bare ground, and buildings within the watershed. Three study sites designated in a separate ecological study within the region were selected as locations to conduct a land cover summary and analysis in this study. The three study sites consisted of a reference site, where the landscape had not been repurposed in any way, a degraded site, where the landscape had been repurposed as a ranch, and a treatment site, where the landscape was once a homestead and has since been restored to represent a healthier riparian and forested wetland habitat. The supervised classification resulted in the reference site having the greatest amount of late successional conifer forest cover, while the treatment and degraded site exhibited the greatest amount of mixed conifer-deciduous forest cover. While the practice of creating a supervised image classification was refined throughout this project, several additional improvements to the classification process were noted. The implementation of these improvements is crucial if future studies using a supervised image classification analysis within Tenmile Creek are adopted into the ecological study being conducted within the region.

Introduction

Tenmile Creek Oregon is located eleven kilometers south of the central Oregon coastal town of Yachats, Oregon (44.221162, -124.094811). Since the forced transfer of property from the Siuslaw natives that relied on the natural goods and ecosystem services that the creek provided in the late 1800s, migrants from around the United States as well as many other countries traveled to the Pacific Northwest to settle in the central Oregon coast and acquire land from the U.S federal government. Following the 1906 Forest Homestead Act, immigrants purchased land to establish timber harvesting operations, homesteads for ranching, and private residential properties within Tenmile Creek (Radtke 2017). This transformed the historical landscape from an interconnected ecosystem of riparian areas, wetlands, forests of various successional periods, to a present day fragmented and disturbed watershed.

As part of a larger ecological study that is being conducted in Tenmile Creek, a collaboration between the Environmental Science Graduate Program at Oregon State University, Siuslaw National Forest, Portland Audubon Society, and The Worthy Garden Club was established. This allowed for the effects of restoration, preservation, landscape change, and ecological differences within three independent sites in Tenmile Creek to be possible, and for the collection of hydrological, soil, vegetation, avian, and carbon storage field data within the watershed. Through this partnership, three properties were available for study in Tenmile Creek, all of which reside within ten kilometers of each other along the main stream of the watershed. While all three sites are within the same ecosystem, each site's history has affected the status and health of the habitats within the three properties.

The Siuslaw National Forest owned parcel of land near Tenmile Creek itself has remained untouched and preserved throughout the past two centuries, resulting in an old growth forest with pockets of riparian wetlands. The Worthy Garden Club property was a ranch for several years, and has been logged historically, resulting in a fragmented and unhealthy habitat that resembles very little characteristics in comparison to unaltered habitats in the same ecosystem. Finally, the Audubon plot was once a homestead that had many anthropogenic changes and has since undergone several waves of restoration efforts within the past three decades to restore the habitat to a similar state as the habitats found in unaltered regions in the same ecosystem. All three sites were categorized by their ecological condition, deeming the Siuslaw National Forest parcel, the Worthy Garden Club parcel, and the Audubon parcel as reference, degraded, and treatment sites respectively.

In the interest of expanding the methods of which habitat summaries and comparisons of the region can be made remotely, additional knowledge and practice in remote sensing analysis was necessary. Learning how to conduct a supervised image classification in Tenmile Creek would satisfy three goals:

1. To advance and refine my remote sensing skills over the course of the academic quarter in hopes of using these practices in a peer reviewed setting
2. To evaluate the credibility and usefulness of a supervised image classification within my study sites and make recommendations on how it could improve in future studies
3. To summarize and compare the predominant cover type in the reference, degraded, and treatment sites in Tenmile Creek

Methods

Using images collected by the National Agriculture Imagery Program (NAIP) (NAIP 2020), several attempts in generating a supervised classification within the boundary of the Tenmile Creek watershed in the central Oregon coast were made. Supervised classifications were generated using ArcGIS Pro (v.3.0.2) k-nearest neighbor classifying strategy from training samples within the Tenmile Creek watershed. Three aerial images that encompass the boundary of the Tenmile Creek Watershed were used for the classification. Each raster file consisted of a red, green, blue, and near-infrared band, and was blended into one continuous raster within the confines of the watershed boundary. This was done to compensate for slight differences in times the aerial images were taken (2-3 hours) and to restrict all training areas and regions that the classification would take place in within the watershed that the study sites were located. The supervised image classification was organized into six different categories: buildings, bare ground, late successional conifer forest, mixed conifer-deciduous forest, deciduous forest, and conifer forest. Finally, the percentage of cover that each of the six categories generated from the supervised image classification were summarized across the three, ten-hectare study sites.

Due to the three study sites (reference, degraded, and treatment) each being 10 hectares and the Tenmile Creek watershed spanning across 6059 hectares, two methods for generating training areas were used. Locations that had been visited in abundance during the data collection portion of the ecological study in Tenmile Creek were prioritized as locations where training sample polygons would be created for the supervised image classification to reference. For example, if a large portion of the reference site consisted of late successional conifers in the upper canopy during the 2022 field season, it was likely that the aerial image taken in that location in 2020 would resemble the same grouping of vegetation and could be used as an accurate training area once cross referenced and verified with ground control points collected during the field season. However, due to the large area that the Tenmile Creek watershed encompasses and the limited accessibility and additional cost of visiting locations outside of the study sites created in the ecological study, this method yielded a small amount of training samples. Additional training samples were created further from the study sites by referencing other sources of geospatial data contributed by a variety of groups and individuals. For example, the United States Forest Service has several maps depicting the locations of timber operations within the central Oregon coast (United States Department of Agriculture, Forest Service 2022). The combination of these two training area sample methods allowed for accurate training samples to be created across a wide arrangement of locations within the watershed.

Results

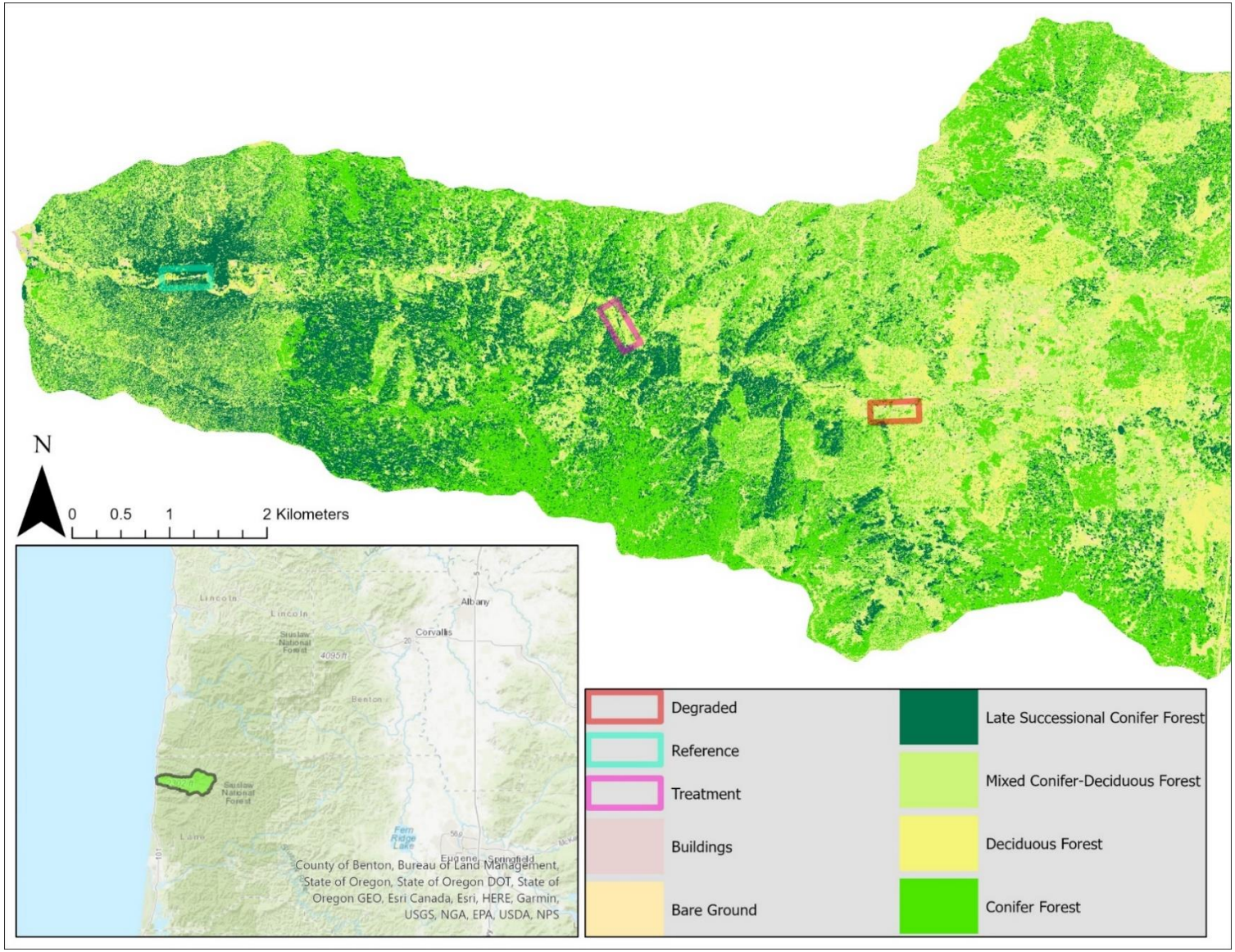
The supervised image classification (Map One) exhibited an overall accuracy of 57% when assigning pixels to each of the six classes. The category with the greatest amount of error of commission (type one error) was the mixed conifer-deciduous forest category with a user accuracy of 44.12% while the category with the least amount of error of commission was the bare ground category with a user accuracy of 77.67%. The category with the greatest amount of error of omission (type two error) was the mixed conifer-deciduous forest category with a producer accuracy of 49.07% while the category with the least amount of error of omission was the late successional conifer forest category with a producer accuracy of 89.72% (Table One).

The reference site within Tenmile Creek had the most late successional conifer forest cover by pixel compared to all five other classification cover types, followed by deciduous forest and mixed

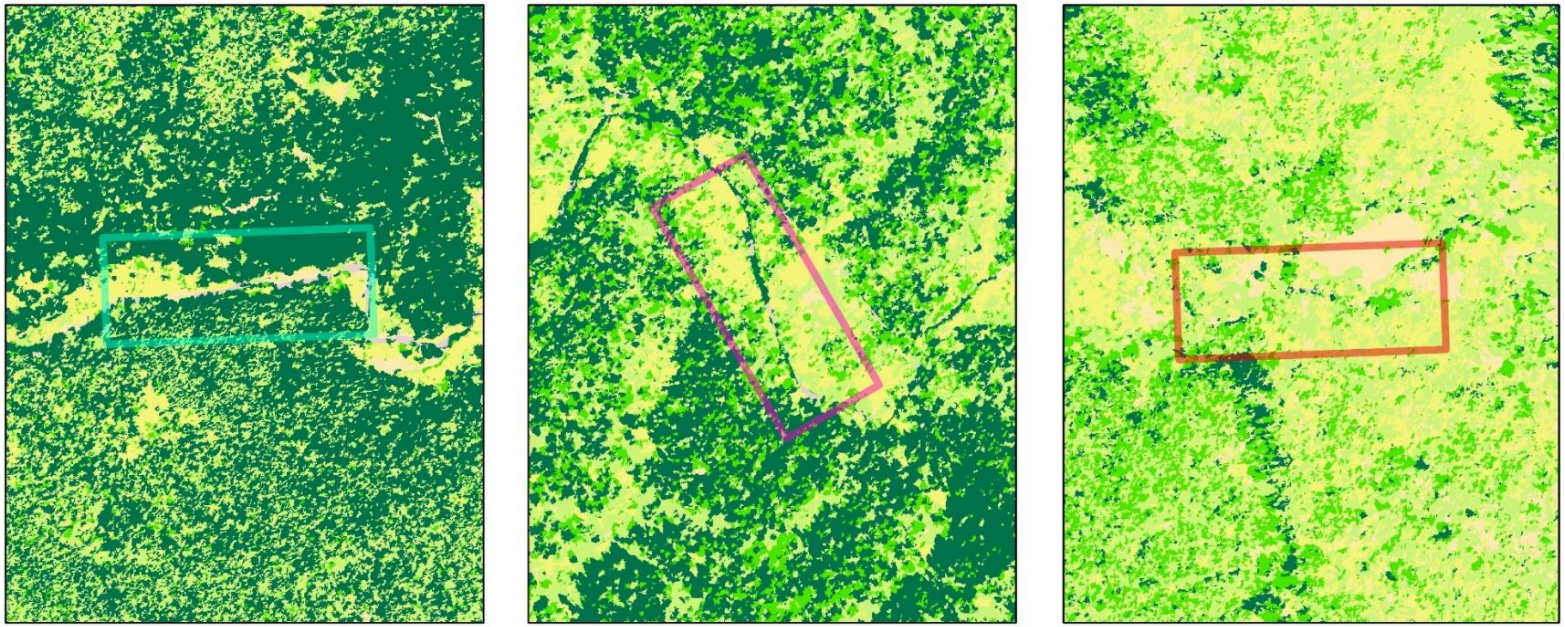
conifer-deciduous forest cover. The treatment and degraded sites both had mixed conifer-deciduous forest as the most predominant cover type, followed by deciduous forest. However, the treatment site had conifer forest cover as the third most predominant form of cover while the degraded site has bare ground as the third most predominant cover type. Finally, the reference and degraded sites demonstrated somewhat of an inverse relationship to one another, with a large cover percent of late successional forest cover and a small cover percent of bare ground cover in the reference site while the opposite was found in the degraded site (Map Two).

	True Values									
Predicted Values	ClassValue	Buildings	Bareground	Late Successional Conifer Forest	Mixed Conifer-Deciduous Forest	Deciduous Forest	Conifer Forest	Total	User Accuracy (Type I Error)	Kappa (Total Accuracy)
	Buildings	182	23	0	0	0	0	262	69.47%	
	Bareground	19	153	0	25	0	0	197	77.66%	
	Late Successional Conifer Forest	0	0	192	8	0	26	356	53.93%	
	Mixed Conifer-Deciduous Forest	13	36	6	105	29	45	238	44.12%	
	Deciduous Forest	0	2	0	54	185	7	248	74.60%	
	Conifer Forest	0	0	16	22	0	136	197	69.04%	
	Total	214	214	214	214	214	214	1498	-	
	Producer's Accuracy (Type II Error)	85.05%	71.50%	89.72%	49.07%	86.45%	63.55%	-	63.62%	
	Kappa (Total Accuracy)									**57.55%**

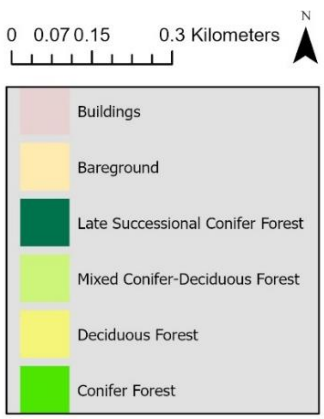
Table One: *An accuracy assessment matrix (error matrix) of all categories of the supervised image classification of Tenmile Creek. Values within the class value portion of the matrix represent the ground truth values of the image classification while values in percentages represent the accuracy of the classification itself in user and producer accuracy calculations. The kappa percentage value represents the overall accuracy of the supervised image classification.*

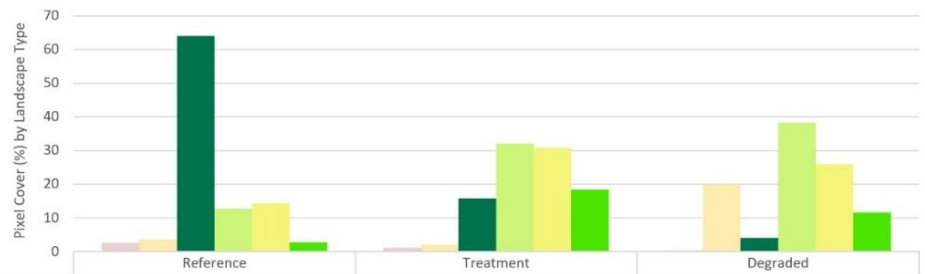


Map One: *A supervised image classification of the Tenmile Creek Watershed along the central Oregon coastline. The three study sites in the ecological study are highlighted within the watershed of Tenmile Creek.*



Reference Site in Tenmile Creek | Treatment Site in Tenmile Creek | Degraded Site in Tenmile Creek





	Reference	Treatment	Degraded
Buildings	2.67	1.09	0.35
Bare Ground	3.52	1.95	19.87
Late Successional Conifer Forest	64.02	15.78	4.04
Mixed Conifer-Deciduous Forest	12.69	32.09	38.26
Deciduous Forest	14.28	30.75	25.92
Conifer Forest	2.81	18.34	11.56

Map Two: *A comparison of the three study sites within Tenmile Creek by cover type generated from the supervised image classification followed by a summary of the percentage of cover each image classification category represents within the ten-hectare study areas in the watershed.*

Discussion

As mentioned previously, many attempts at creating a supervised image classification of the Tenmile Creek watershed were made for this study. The final supervised image classification output was not found to be accurate in categorizing pixels by their assigned class from the training samples supplied to it, with an over accuracy percentage of 57.55%. There are several issues that were identified following the image classification output that lead to the supervised image classifications abysmal accuracy.

The training samples used in the supervised image classification was unnecessarily grouped into redundant categories of classification from an ecological perspective. For example, the decision to create classes of deciduous, conifer, and mixed conifer-deciduous forests was made a priori in an attempt to identify portions of the watershed that had a mix of hard and softwoods. However, given the size of the watershed and the large number of pixels within the images used for the classification, having a mixed conifer-deciduous forest category was redundant, for a section of land within the watershed that has a forest of various soft and hard woods would have a mix of conifer and deciduous forest pixels in it. Furthermore, the mixed conifer-deciduous forest category had the lowest producer's and user accuracy rating, leading to a lower overall accuracy rating for the supervised image classification. Future image classifications within Tenmile Creek should prioritize the use of simple and independent categories, such as conifer forest, deciduous forest, or even classifications by individual species of tree, and omit mixed categories such as mixed conifer-deciduous forest.

To compensate further for the abysmal accuracy rating of the supervised image classification, future classifications of the Tenmile Creek region should prioritize the use of aerial images containing a greater number of bands than the images supplied by the National Agriculture Imagery Program (NAIP) for additional information regarding a categories spectral signature. In early iterations of the Tenmile Creek supervised classification, when creating categories for the training samples used, two additional categories named water and invasive species were created and supplied training samples. However, due time constraints these categories were omitted from the final image classification, since their spectral signatures were like other objects or landscapes in Tenmile Creek, leading to the supervised image classification incorrectly assigning pixels to categories. The most common occurrence of this was present in water being assigned to the buildings category, or deciduous forests being labeled as invasive species, and vice versa. The decision to omit the water and invasive species category from the final supervised image classification was made to limit the amount of incorrect pixels throughout the Tenmile Creek watershed, since the image classification output had designated several regions as having water and/or invasive species in it that are known to not have either. Unfortunately, the lack of a water category led to many locations within the study sites to be incorrectly labeled as buildings or bare ground pixels, further convoluting the results of the cover distribution within each study site. Aerial images taken with additional bands would be able to produce a more accurate supervised image classification due to the increased amount of information of each category's spectral signature when assigning training samples and should be used in future supervised classifications of the Tenmile Creek watershed.

Finally, this study showed that an ecological assessment of the differences in cover types between the reference, degraded, and treatment sites in the study area would be possible following the development of an additional supervised image classification in the Tenmile Creek watershed. If improvements are made to the training samples, category assignments, and aerial image quality

components of a supervised image classification creation in Tenmile Creek, a statistical analysis of both the differences in cover type between each study site, and a temporal analysis of the changes that occur in each study site are feasible. The development of ecological research questions is necessary to conduct a statistical test of some sort and should be developed simultaneously with the creation and refinement of a more accurate supervised image classification within the Tenmile Creek watershed.

Conclusion

Supervised image classifications within Tenmile Creek were possible given the amount of information and images that have been collected within the previous decades. However, the supervised image produced in this study had an abysmal overall accuracy rating due to limitations in study design, training sample classifications, aerial image quality, and time constraints. The results of this study highlight the potential for future remote sensing and ecological studies to be conducted within Tenmile Creek, given that the suggested recommendations made in this study regarding supervised image classification creation are used.

References

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